

A Thyroid Nodule

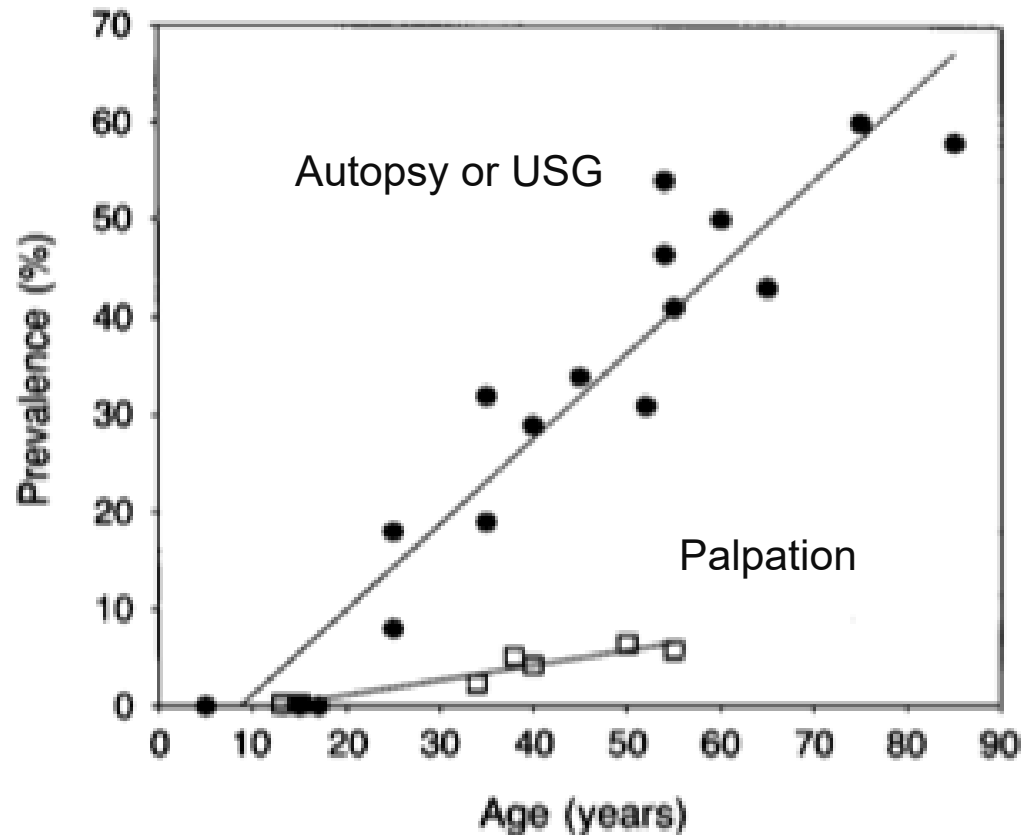
Benign thyroid nodules

Thyroid cancer

Thyroid nodule

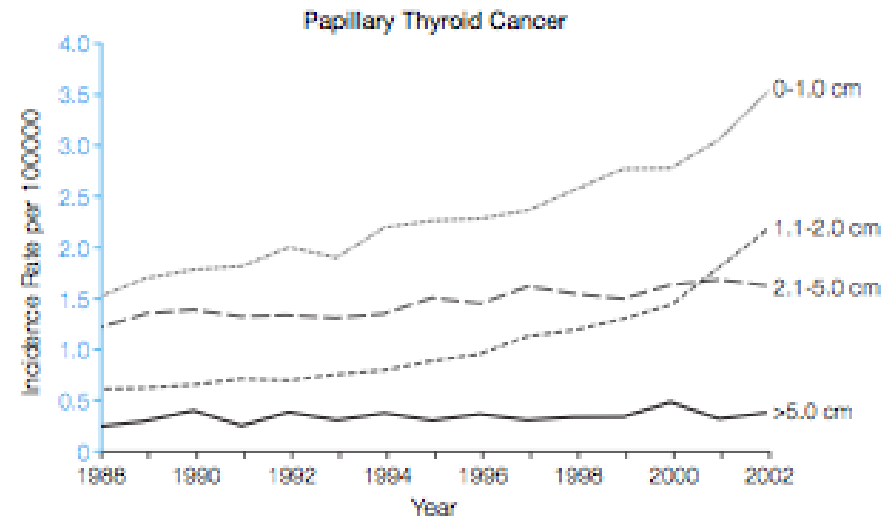
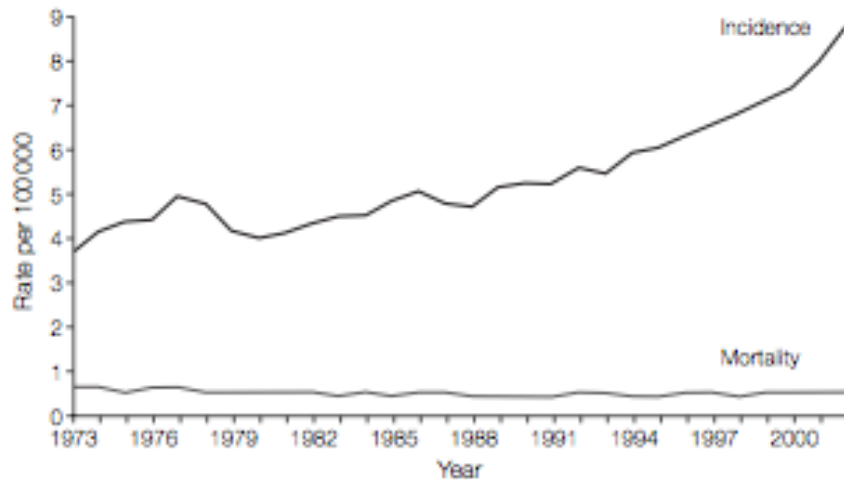
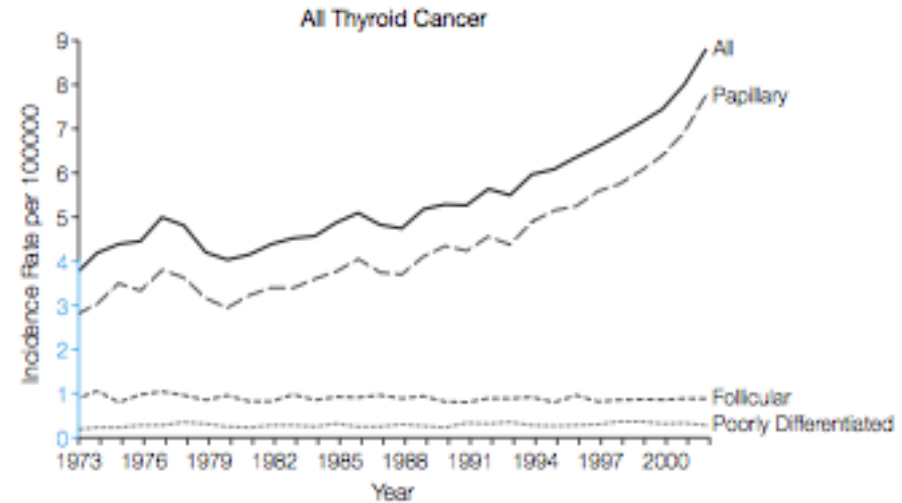
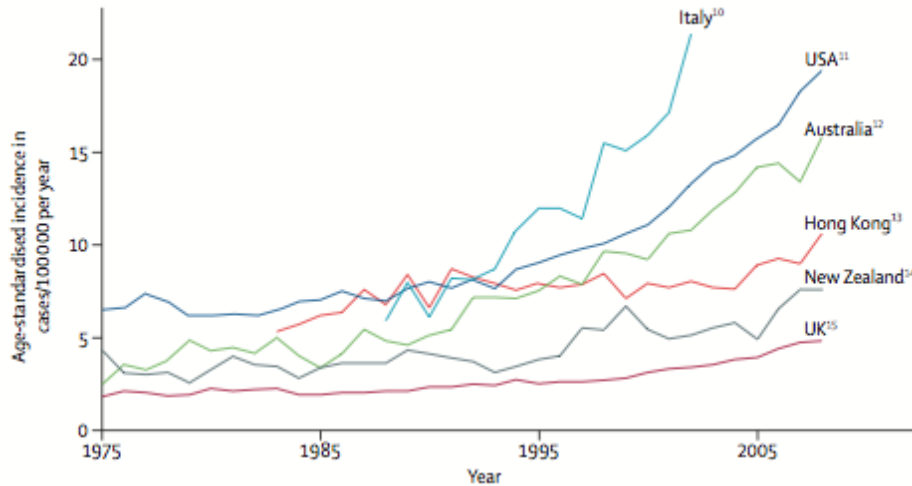
Prevalence

- Extraordinary common
- Depends on:
 - iodine deficiency
 - ionizing radiation
 - gender
 - age
 - method of detection



Thyroid cancer

Incidence, mortality and stage



Goitre

Classification

- Simple goitre (endemic or sporadic)
 - diffuse
 - nodular
- Neoplastic goitre
 - benign
 - malignant
- Toxic goitre
 - diffuse toxic (Graves')
 - toxic nodular (Plummer's)
 - toxic/functioning adenoma
- Thyroiditis
 - bacterial (acute suppurative)
 - viral (subacute)
 - lymphocytic/Hashimoto/autoimmune (chronic)

Thyroid nodule

Pathology

- Nodular goitre : colloid / haemorrhagic cystic / complex/ hyperplastic / adenomatous nodule (70%)
- Benign follicular adenoma: mainly non-toxic (15%)
- Well-differentiated thyroid carcinoma (10%)
- Miscellaneous: other thyroid malignancies, thyroiditis (5%)

Thyroid nodule

Symptoms/Presentations

- Neck swelling/mass
- Pain or discomfort
- Local pressure symptoms/voice
- Thyroid dysfunction
- Incidental or unrelated:
 - physical examination
 - imaging

Thyroid nodule

Investigations

- Blood tests: TSH+free T4
- Ultrasound
- FNAC (+molecular testing)

- ESR, thyroid antibodies, calcitonin, genetic testing...
- Imaging: radioisotope scan, CT scan/MRI, PET scan
- Endoscopy

- Thyroidectomy: diagnostic+therapeutic

Investigation

Ultrasonography (USG)

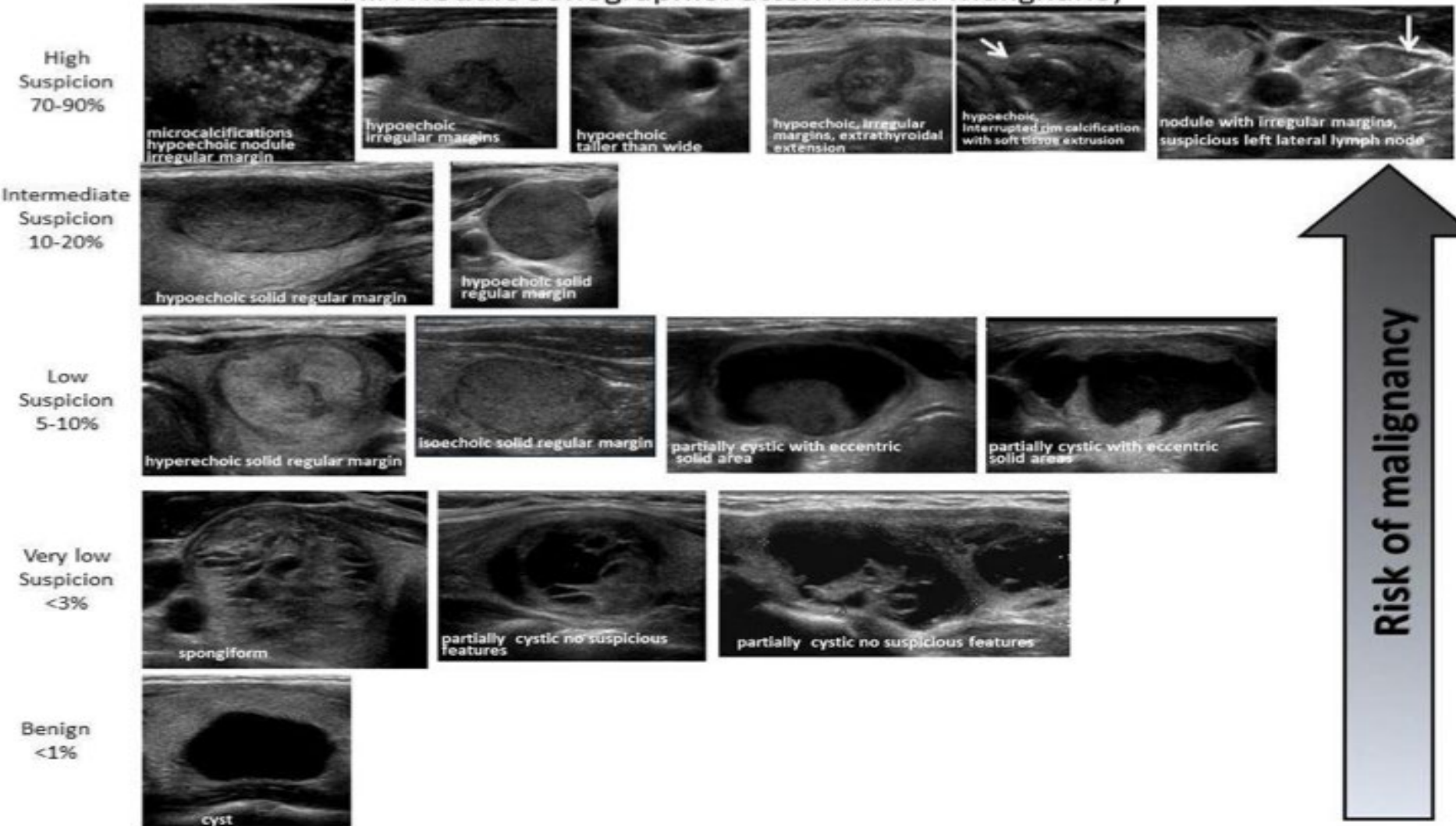
- B-mode real-time
- Non-invasive, no radiation, convenient and cheap
- Highly sensitivity but relatively low specificity
- Role:
 - extend physical examination
 - select nodules for FNAC
 - guide needle aspiration
 - for all patients with goitre/palpable nodule
 - not recommended as a screening test



Thyroid nodule

USG assessment

ATA Nodule Sonographic Pattern Risk of Malignancy



Thyroid Imaging Reporting and Data Systems (TIRAD)

Thyroid nodule

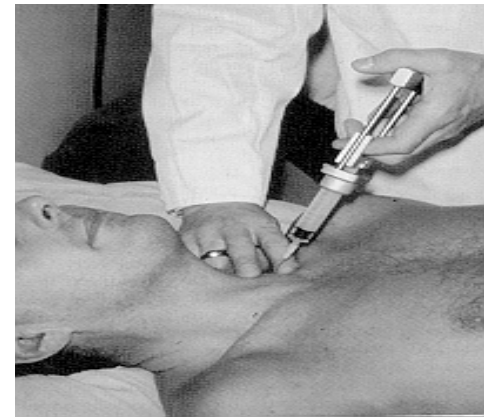
Fine needle aspiration cytology (FNAC)

Bethesda classification

Diagnostic category	Risk of malignancy	Usual management
I. Non-diagnostic	1-4	Repeat FNA
II. Benign	0-3	Clinical follow-up
III. AUS or FLUS	5-15	Repeat FNA
IV. Follicular neoplasm	15-30	Surgical lobectomy
V. Suspicious of malignancy	60-75	Surgical lobectomy(FS)+TT
VI. Malignant	97-99	Total thyroidectomy (TT)

AUS: atypia of undetermined significance

FLUS: follicular lesion of undetermined significance



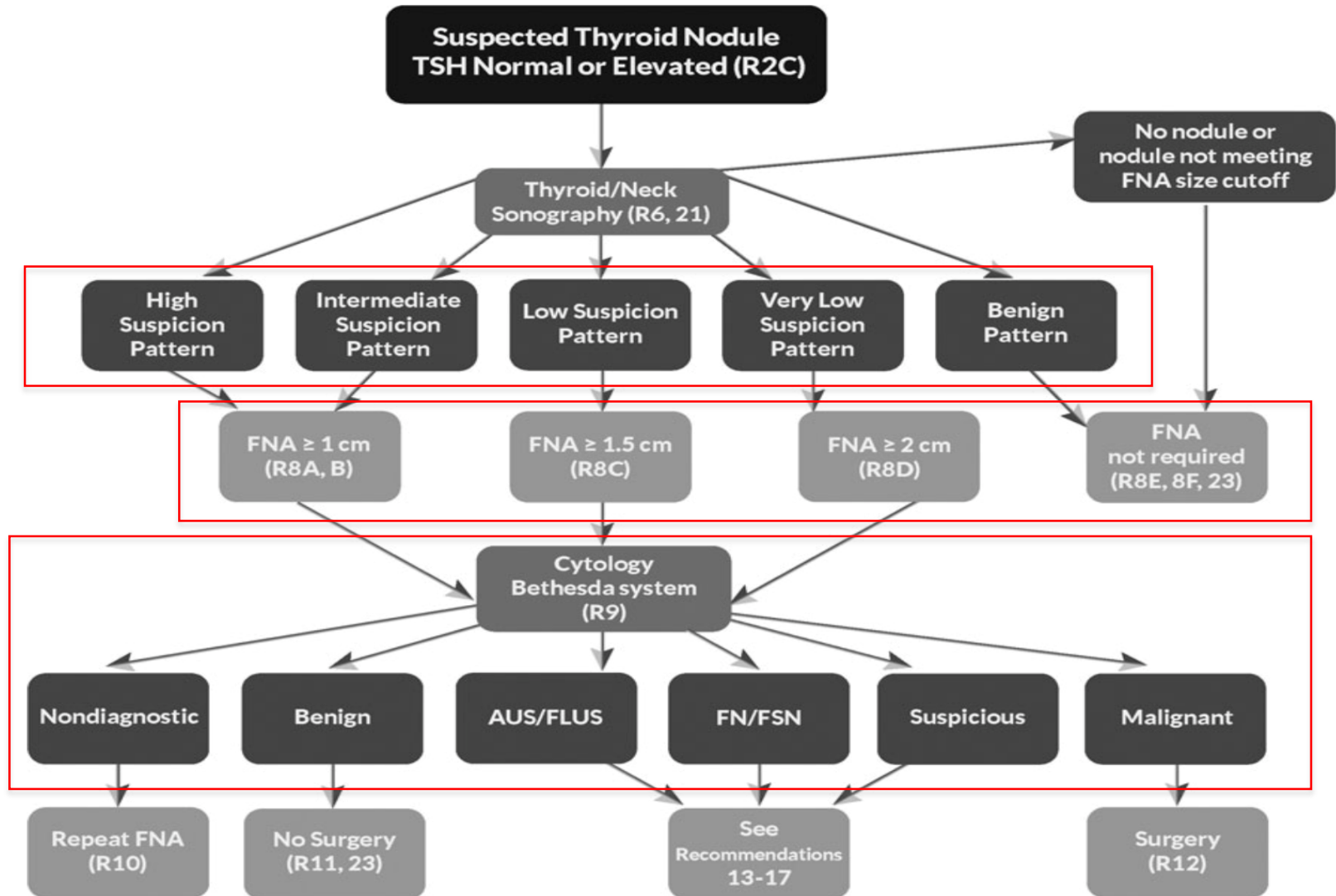
FNA

Molecular testing

- Somatic mutations
 - BRAF: 40-64%
 - panel of mutations (Asuragen miRInformTM): 63-80%
 - not sensitive but specific: high PPV (rule-in test)
- Gene expression evaluation and miRNA classifier
 - ThyroSeq v3 multigene genomic classifier (DNA-RNA)
 - Afirma[®] genomic sequencing classifier (RNA)
 - highly sensitivity but not specific with high NPV (rule-out test)
 - for indeterminate (CAT III-IV: FLUS/FN) nodule
 - avoid diagnostic thyroidectomy in 50%

ATA guidelines

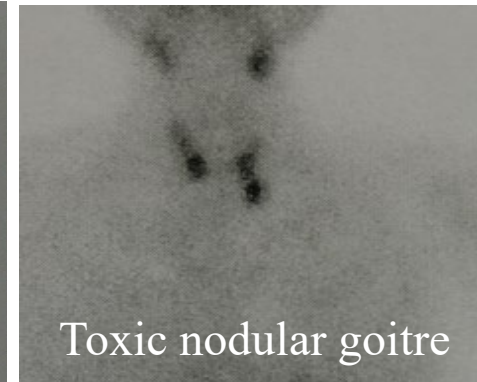
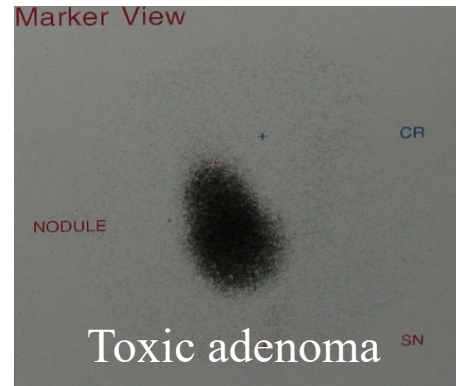
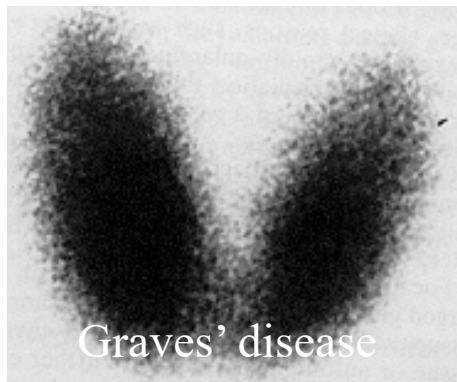
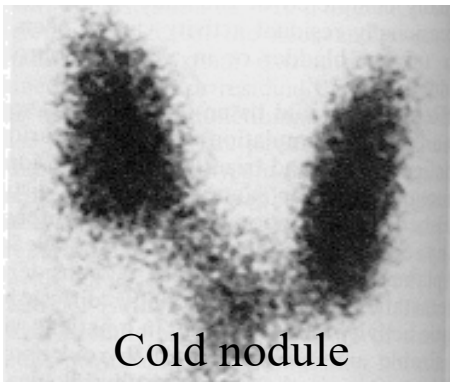
Management of thyroid nodule



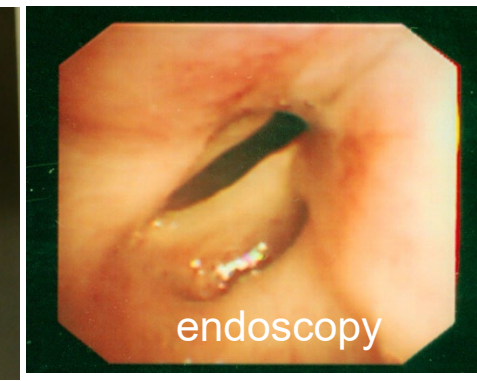
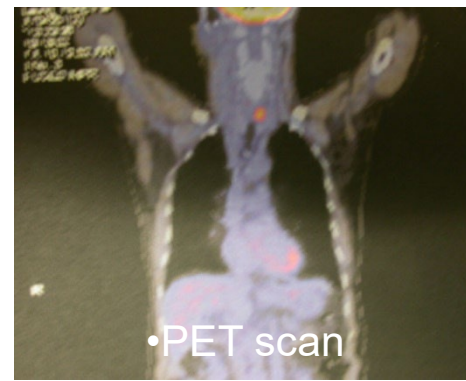
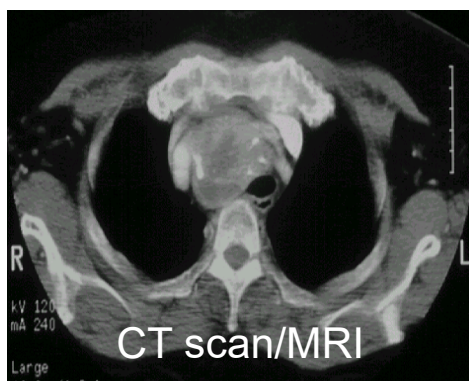
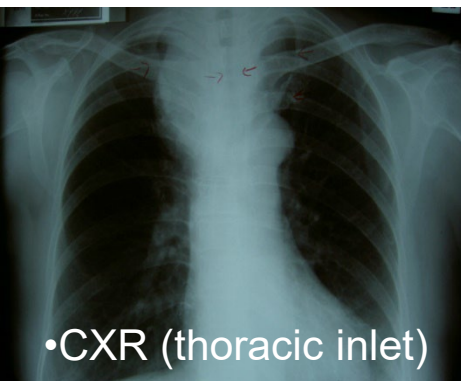
Other investigations

Imaging and endoscopy

- Radio-isotope scintigraphy (I^{123} or Tc^{99m})
 - diagnosis of malignancy: low sensitivity and specificity
 - functional assessment in thyrotoxic patients



- Others: CXR, CT/MRI, PET scan and endoscopy



Benign thyroid nodules

Indications of treatment

- Symptomatic (size of goitre/nodule)
- Increase in goitre size
- Trachea compression or deviation
- Retrosternal extension
- Suspected malignancy
- Cosmetic considerations/patient wish

Benign thyroid nodules

Surgical treatment

- Unilateral lobectomy (hemithyroidectomy)
 - for uninodular goitre
 - safe, minimal morbidities, diagnosis and cure
 - future reoperation on contralateral lobe without difficulty
 - around 10-20% chance of hypothyroidism
- Total/near-total thyroidectomy (bilateral thyroidectomy)
 - for symptomatic multinodular goitre
 - no recurrence/need of reoperation
 - additional surgical risk (hypoparathyroidism)
 - needs long-term thyroxine replacement
- Partial or bilateral subtotal thyroidectomy
 - rarely indicated

Thyroid nodule

Nonsurgical ablative treatment

- Ethanol injection (PEI) Cytoplasmic protein dehydration/coagulation necrosis
Cystic/predominantly cystic
- High-intensity focused ultrasound (HIFU)
- Radiofrequency ablation (RFA) Thermal coagulation necrosis
- Percutaneous laser ablation (LA) Complex or solid nodule
- Microwave ablation (MWA)



Thyroid nodule

Nonsurgical ablative treatment

Table 2. Minimally Invasive Techniques for Treatment of Symptomatic, Benign Thyroid Nodules

Method	Mechanisms	Candidate Nodules	Expected Volume Reduction	Adverse Effects	Cost Estimate
Percutaneous ethanol injection (PEI)	Dehydration of cytoplasmic proteins, coagulation necrosis, and fibrosis	Cystic or predominantly cystic nodules	~60%	• Pain • Burning sensation • Hematoma • Dyspnea • Voice change	Simplest, least expensive option (\$50-\$100)
Radiofrequency ablation (RFA) ^a (101, 103, 104)	Thermal coagulation necrosis	Mixed or solid nodules	47.7%-96.9%	Overall complication rate 2.11% (major complications 1.27%) Major: • Voice change • Nodule rupture • Hypothyroidism • Brachial plexus injury Minor: • Pain • Thermal propagation outside of the thyroid • Fever • Skin burns • Hematoma • Transient hyperthyroidism/transient thyroiditis	Equipment \$25,000; \$800 per session
Laser ablation (LA) ^a (101, 102)	Thermal coagulation necrosis	Mixed or solid, functional, or nonfunctional nodules	62 ± 22% ^b	• Pain (10%) • Fever (8%) • Vasovagal reaction (1%) • Voice change (0.5%) • Hematoma (0.4%) • Skin burn (0.1%)	Equipment with built-in laser source: \$12,000 Nd:YAG laser source: \$15,000-\$20,000 \$500 per session
Microwave ablation (MWA) (105-107)	Thermal coagulation necrosis	Mixed or solid nodules	50%-70% depending on nodule composition (solid require more energy than cystic nodules)	• Pain (25%) • Transient voice change (1%) • Hematomas • Burns (2/30) • Horner syndrome (1/30)	Equipment: \$35,000 \$400 per session
High-intensity focused ultrasound (HIFU) (98-100)	Thermal coagulation necrosis	Mixed or solid nodules	49%-69%	• Hypothyroidism (1.4%-2.3%) • Hoarseness • Neck swelling	Focused thermal tissue destruction without needles; Equipment: \$400,000 \$350 per session

Thyroid nodule

Nonsurgical ablative treatment

- Volume reduction rate (VRR: 50-80%)
- Patients' selection:
 - Symptomatic/single/hyperfunctioning nodule
 - <4-cm or 20mL/growth in sonogram
 - benign (FNACx2)
 - motivated patients
- Expectations: preserve function, minimally invasive, avoiding GA, day procedure, scarless, cosmetic outcome.....
- Limitations: not cure, unsatisfactory shrinkage, regrowth, additional procedure, risk unique to ablation, expense.....

Thyroid carcinoma

Major histologic types

- Well-differentiated (90%)
 - papillary (90%)
 - follicular (10%)
- Poorly differentiated (insular) (<5%)
- Undifferentiated (anaplastic) (<5%)
- Medullary carcinoma (<5%)
- Others (1%): lymphoma, squamous cell carcinoma, sarcoma, metastasis ...

Well-differentiated thyroid carcinoma

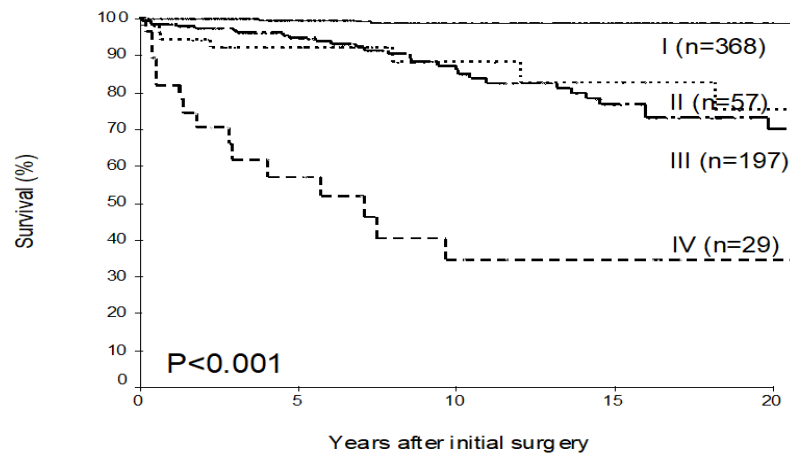
Management considerations

- Extent of thyroidectomy
 - hemithyroidectomy vs total (bilateral) thyroidectomy
- Nature/aim and extent of lymph node/neck dissection
 - prophylactic or therapeutic
 - central and/or lateral compartments
- Postoperative adjuvant therapies
 - radioiodine (I^{131}) ablation
 - external beam irradiation
 - thyroxine (T_4) suppressive therapy

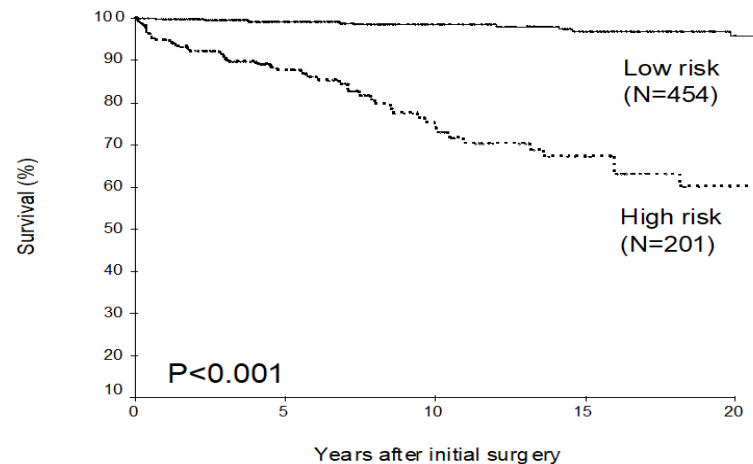
Well-differentiated thyroid carcinoma

Risk groups

	Low risk	High risk
% of patients	85-90	10-15
Mortality (10-20 yrs)	2-5%	40-50%
Recurrence	10%	45%



AJCC pTNM staging



AMES (Age, metastasis, extent and size) stratification

Papillary thyroid carcinoma (PTC)

Management controversy for low-risk patients

- Early stage: <4 cm, no invasion, no LN metastasis
- Majority as papillary microcarcinomas (PTMC): <1-cm
- Survival: nearly 100%

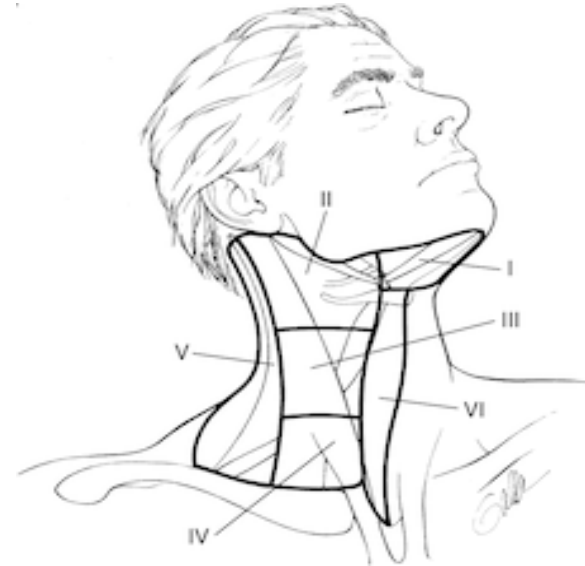
Total thyroidectomy	Hemithyroidectomy
Commonly multifocal and bilateral	Excellent survival and low recurrence
Allows RAI ablation and thyroglobulin (Tg) monitoring	Avoid T4 replacement
Low morbidity rate by experienced surgeons	Lower morbidity

- Overtreatment/surgical risk vs avoid recurrence/facilitate followup
- Patients' vs physicians' preference
- Personalized treatment

Cervical lymph nodes in PTC

Management

- Central (level VI: pre- and para-tracheal): same incision
 - prophylactic/routine vs therapeutic
 - reduction in disease recurrence/reoperations
 - upstaging disease and increased use of I¹³¹
 - increased in morbidity: hypoparathyroidism
- Lateral (level II-V): extent incision
 - usually therapeutic
 - clinically involved or FNAC proven nodes
 - excisional (Berry-picking)
 - modified/functional/selective neck dissection



Adjuvant therapy

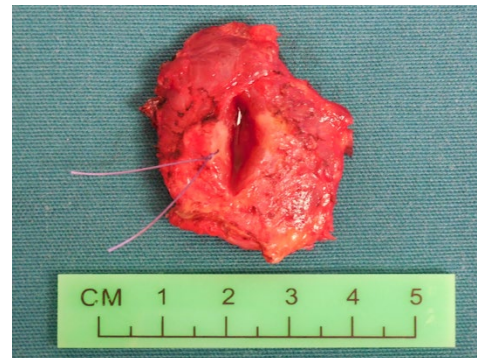
Use of radioiodine (I^{131}) ablation

- Follow total/near-total thyroidectomy
- Thyroxine withdrawal vs thyrogen (recombinant TSH)
- Therapy: high risk (stage III/IV disease)
 - gross persistent/residual disease
 - nodal or distant metastases
- Ablation: low-risk (selected stage I and stage II)
 - destroy occult microscopic foci
 - facilitate disease surveillance
 - Low dose (30mCi) as effective as higher dose (100 mCi)
 - short and long-term adverse effects

Papillary thyroid microcarcinoma (PTMC)

Definition and relevance

- WHO: <1-cm+detected incidentally
- AJCC pTNM staging
 - <1-cm
 - majority pT1a or stage I disease
 - usually incidentally detected
- low-risk or extremely low-risk
- excellent or almost 100% survival
- less aggressive treatment (lobectomy) or active observation



PTMC

Active observation

- Biopsy proven incidental PTMC
- No metastases/invasion/high grade subtype
- Tumour progression (up to 20 years)
 - ↑ size (3mm): 8%; novel nodal metastases: 3.5%
- Surgical treatment for tumour progression
- Feasible treatment option

Ito Y & Miyauchi A. Thyroid 2003; JCEM 2007; World J Surg 2010; Curr Opin Oncol 2015; Sugitani I al World J Surg 2010; Miyauchi A. World J Surg 2016; Thyroid 2017; Morinelo E, et al JCEM 2019

Thermal ablation

- Meta-analysis and propensity matched analysis
- Complete ablation/volume reduction/safety
- Tumour progression, recurrence, nodal metastasis same
- Possibly another alternative for PTMC without metastasis

Choi YS, et al Thyroid 2020; Cho, SJ, et al Thyroid 2020; Yan L, et al. Thyroid 2021

Thyroid carcinoma

Medullary thyroid carcinoma (MTC)

- Calcitonin (basal/stimulated) as marker
- Total thyroidectomy
- Central+lateral neck dissection
- Familial cases (1/3)
 - MEN IIA
 - MEN IIB
 - familial MTC
- Genetic analysis: RET proto-oncogene
 - prophylactic thyroidectomy (5-10 yrs)



Thyroid carcinoma

Anaplastic carcinoma

- Elderly with poor co-morbid state
- Aggressive and rapidly growing
- Locally advanced disease
- Frequent distant metastases
- Lack of effective treatment
- Chemoirradiation+resection+target Rx
- Invariable palliative and fatal

